# Eduardo M. Ramírez

Software & Data Engineer · Resistencia, Chaco, Argentina

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## SUMMARY

Software & Data Engineer with 3+ years in charge of the systems of BIAMAQ, an industrial machinery company with 3 branches in Argentina. I design, build and run end to end the software the company uses every day: rental and after-sales management, a sales assistant with product recommendations, OCR/LLM invoice automation, ETL and a data warehouse on top of the ERP, BI for management, and the infrastructure with Docker and CI/CD. I work with Python, Django REST Framework, SQL Server, PostgreSQL, React and TypeScript. Final-year Information Systems Engineering student at UTN, with an exchange semester in CESI's Data Science master's programme in France.

## EXPERIENCE

**Software & Data Engineer · IT Lead — BIAMAQ SRL** Nov 2023 – Present

Industrial machinery, 3 branches · Full-time · Remote, Argentina

- Led the re-engineering of the rental module into RentalIA, replacing paper contracts and fragmented legacy workflows with tablet-signed PDFs, returns, workshop and ARCA e-invoicing, synced with the ERP or standalone.
- Transformed BIAMAQ's sales workflow with customer segmentation, product recommendations and commercial follow-up, integrating the ERP and data warehouse with nightly retraining: its suggestions are right twice as often as suggesting best sellers (~40% vs ~20% in the top 3, offline).
- Automated invoice, expense and product-data workflows with OCR/LLM assistants and tax checks, reducing estimated manual processing time from 8 to 2 hours (6 hours saved; 75% less manual time).
- Built BIAMAQ Reports, the BI platform management uses daily: sales, purchases vs sales, 80/20 customers, receivables aging, seller orders and workshop stock. Also maintain Power BI dashboards and daily replenishment metrics.
- Built ETL pipelines that load the SQL Server ERP into a data warehouse (sales, purchases, stock, receivables, service and rentals), with daily loads and monthly snapshots scheduled with Celery.
- Built Service BIAMAQ: versioned repair quotes, workshop tracking, ERP orders and Infolab history.
- Run the infrastructure: Docker on Linux servers and CI/CD with GitHub Actions on a self-hosted runner.

Django REST Framework · PostgreSQL · SQL Server · React · Playwright · Docker · Python · Pandas · scikit-learn · TypeScript · OCR · LLM · Power BI · Celery · Redis · Linux

## SELECTED PROJECTS

**Malaca** — Client project · E-commerce

Ecommerce with guest checkout, Mercado Pago, stock reservations, shipping, order tracking and an admin panel.

Django REST Framework · PostgreSQL · React · TypeScript · Vite · Cloudinary

**Casa Palma** — Client project · Business management

In development. Management system for a collaborative store of local makers: sales register with mixed payments, stock per maker, expenses, payouts, Excel and Tiendanube import, and role-based permissions.

Next.js · TypeScript · PostgreSQL · Drizzle ORM · Vitest

**Chedul** — Personal project · Team of 3 chedul.vercel.app

Degree planner for UTN students. Led the migration from Django to a Go REST API and a React + TypeScript SPA, with Testcontainers integration tests; deployed on Cloud Run and Vercel.

Go · React · TypeScript · Vite · PostgreSQL · Docker

**Last.fm Data Platform** — Personal project · Data Engineering github.com/deadour/lastfm-data-platform

Data platform over ~117k listening events: incremental API ingestion with resumable backfills, Bronze/Silver/Gold layers in Parquet, MusicBrainz enrichment and a Streamlit dashboard.

Python · Parquet · Last.fm API · MusicBrainz · Streamlit · pytest

**Dynamo** — Personal project · Full-stack web app dynamo-1.onrender.com

Workout tracker: Django REST API split into domain apps with per-user data isolation covered by tests, metrics derived from sets, React + TypeScript frontend and CI on GitHub Actions.

Django · Django REST Framework · PostgreSQL · React · TypeScript · Docker

**Diabetes Prediction with Deep Learning** — Academic project · CESI France · Team of 4

End-to-end ML pipeline on a highly imbalanced health dataset: PyTorch model, oversampling and threshold tuning to cut false negatives, SHAP/LIME explainability and a FastAPI prototype.

Python · PyTorch · Keras · NumPy · SHAP · LIME

**Classroom Noise Monitor** — Academic project · CESI France · IoT · Team of 4

Real-time classroom noise monitor: ESP32-S3 with a digital microphone, readings sent through ESPHome to Home Assistant, InfluxDB and Grafana dashboards, never recording audio.

ESP32 · ESPHome · Home Assistant · InfluxDB · Grafana

## EDUCATION

**Universidad Tecnológica Nacional (UTN FRRe)** — Resistencia, Argentina
Information Systems Engineering · Final year in progress

**CESI École d'Ingénieurs** — Rouen, France
International academic exchange · Data Science master's programme · ARFITEC scholarship · Spring semester 2026
One semester of coursework in data science, AI, IoT and web development, with engineering projects in multidisciplinary teams.

## TECHNICAL SKILLS

**Languages:** Python, SQL, TypeScript, JavaScript, Go
**Backend:** Django, Django REST Framework, FastAPI, Celery, Redis, REST APIs
**Data:** SQL Server, PostgreSQL, Pandas, Parquet, ETL, Data warehousing, Power BI, Streamlit
**Machine learning:** scikit-learn, PyTorch, Keras, NumPy, SHAP
**Frontend:** React, Next.js, Vite, Tailwind CSS
**Infrastructure & testing:** Docker, Linux, Nginx, Git, GitHub Actions, CI/CD, Cloud Run, Vercel, pytest, Playwright

## CERTIFICATIONS

Programación Web Full Stack — Argentina Programa, 2023
Inglés Técnico para Desarrolladores de Software — Nivel 6 — Centro Universitario de Idiomas (CUI), 2023
Power BI: Desde cero para principiantes — Udemy, 2024
Introducción a Machine Learning — Informatorio Chaco, 2023
Python sin fronteras: HTML, CSS, Flask y MySQL — Udemy, 2022

## LANGUAGES

Spanish (Native) · English (B1) · French (A2)